Davide Conficconi

Curriculum Vitae

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HIGHLIGHTS

- Assistant Professor (Untenured track) (Ricercatore a tempo determinato tipo A) at Dipartimento di Elettronica, Informazione e Bioingegneria, Politecnico di Milano, since March 2024.
- Adjunct Professor (Professore a Contratto) of
  Spring'23, Master Course Advanced Computer Architecture (200 students, 5 CFU)
  Spring'23, Bachelor, Foundations of Computer Science to Bioengineers (180 students , 7 CFU)
  Fall'23, Bachelor, Applied Computer Science for Communication Designers (70 students, 5 CFU)
- Co-Author of 10 Journal Papers, 7 out of 10 are Q1, in IEEE Transaction on Parallel and Distributed Systems, ACM Transaction on Reconfigurable Technology and Systems, ACM Computing Surveys, IEEE Transaction on Emerging Topics in Computing, Journal of Parallel and Distributed Computing, and IEEE Journal of Biomedical and Health Informatics.
  Top Conference Publications: DAC, CGO, FPGA, ISCAS, IISWC
  In total: 10 Journal papers, 19 conference papers, and 1 book chapter.
- H-index: 8; Total citations: 164 (Google Scholar, January 2024)
  H-index: 7; Total citations: 97 (Scopus, January 2024)
- Research intern at different international companies: IBM Research Zurich 2021-22, Xilinx (now AMD) Research Dublin 2018-19.
- Different research works awarded with Artifacts Badges, Co-Supervisor of winner team for an international FPGA-based design contest '23,'22, best Ph.D. Intern presentation at IBM department symposium, first prize for the intern competition at Xilinx Dublin, first place in the IEEE Lance Stafford Larson award.
- Artifact Evaluation (AE) Chair of RAW'24, RAW'23
  Program Committee FPl'24, FCCM'24, HEART'24, SAC'24, RAW'24, ISPA '23, GLSVLSI'23, RAW'23, RAW'22
  Artifact Evaluation (AE) Committee CGO '24, '23, ASPLOS '23, '22, PLDI'23
- Guest Editor of a Special Issue on "Secure and Efficient Distributed Computation for Emerging Systems on the Edge" (SUNRISE) on Elsevier Journal of Parallel and Distributed Computing (JPDC) 2023
- Presenting the research work at FPGA'21, RAW'23-21-18, and H^2RC, while presenting the internship work at intern competitions and winning the best presentation award.
- Journal (20+ manuscripts) and conferences (25+ manuscripts) reviewer for several venues
- Task leader in the EVEREST European Project.
- Organizing committee of the CPS Summer School 2023.
- Instructor of the *"Hardware and Accelerators: FPGAs for AI"* for Cefriel, and the Creative Lab at the CPS Summer School.
- Volunteering for the IEEE Larson Award Chair
- Producing around 18 open-source research tools
- Teaching assistant for different courses in the field of digital design and computer architecture from 2019.
- Advisor of three Master thesis, and Co-Advisor of 3 Master thesis; supervisor of 35+ master and bachelor projects.

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SHORT BIO

I am an Assistant Professor (RTDA) at Politecnico di Milano where I got my Ph.D. Information Technology (Feb '22). My research interests revolves around the (co-)design of Domain-Specific Computer Architectures and Systems focusing on reconfigurable architectures, especially FPGAs, design methodologies, computer architectures, design automation techniques, and abstraction layers.

RESEARCH INTEREST

My research spans on three topics about conceiving **Domain-Specific Reconfigurable Architectures** and **Systems** to face Moore's Law end and harvest more performance at a reasonable energy budget. I exploit the domain specialization at the software and the hardware levels, where co-design is central, and combine them with the flexibility of reconfigurable computing systems such as FPGAs to create adaptable computing systems. The three pillars I focus on are **design methodologies** for tailoring an architecture to a domain, design **automation** and **generators** to enable fast and iterative hardware deployments, and **usability** layers, such instruction set architectures, compilers, run-time, and APIs. I am also moving my first steps in designing **classical** computing **accelerators** for **quantum computing**.

EDUCATION

2018 – 17 FEB. 2022 **Doctor of Philosophy** Information Technology *Politecnico di Milano*

2015 – 2018 **Master of Science** WITH HONORS Computer and Engineering *Politecnico di Milano*

2012 – 2015 **Bachelor of Science** Computer and Engineering *Politecnico di Milano*

RESEARCH EXPERIENCE

Politecnico di Milano CURRENT FROM MAR 2024 (MILAN)
Assistant Professor (Untenured Track):
Starting from March 2024, I am an assistant professor (untenured track) (Ricercatore a Tempo Determinato di Tipo A) working in the research field of (co-)designing of Domain-Specific systems and Architectures for computations spanning from the embedded to the high-performance computing fields.

Politecnico di Milano MAR 2022 – FEB 2024 (MILAN)
Post-Doctoral Researcher:
Carrying on my research domain-specific architectures and design automation toolchains for reconfigurable computing systems. Moving also part of my research efforts on quantum computing technologies spanning from efficient and accelerated computations to design automation. Working with M. Santambrogio on managing the research efforts in the system architecture area of the NECSTLab.

IBM Research SEPT 2021 – FEB 2022 (ZURICH)
Research Intern:
Working with M.Lantz's group with D. Diamantopoulos on the cloudFPGA system. Mainly contributing to the cFp_Zoo a set of domain-specific accelerators for the hybrid multi-cloud era, and smaller contribution to the cFDK.

Xilinx Research (now AMD) AUG 2018 – FEB 2019 (DUBLIN)
Research Intern:
Working with M. Blott's group with Y. Umuroglu. Contribution to the bit serial inference accelerator BISMO (ACM TRETS'19) and FPGA-tidbits components, both open sourced on github.

UniCredit JAN 2018 – JUL 2018 (MILAN)
Research Intern:
Collaborating with UniCredit's R&D department on a financial application targeting FPGAs, under the supervision of M. Paris.

Oracle Labs JAN 2018 – JUN 2018 (REMOTE - ZURICH)
Research Assistant:
Working as Research Assistant remotely at Oracle Labs under the supervision of D. B. Bartolini on Graph and Data visualization applications.

TEACHING ACTIVITIES WITH PRIMARY RESPONSIBILITY

Politecnico di Milano SEPTEMBER 2023 – SEPTEMBER 2024
Contract Professor:
Instructor for Bachelor Course of *"Informatica Applicata"*, applied CS for data visualization, for Communication Design Bachelor Degree with 70 students. Credits: 5 CFU. Course Evaluation: n.a. .

CPS Summer School MAY 2023 – SEPTEMBER 2023
Organizing Committe and Instructor:
Member of the Organizing Committee and Instructor of the Creative Lab of the Cyber-Physical Systems (CPS) Summer School held in Alghero (ITA).

Politecnico di Milano FEBRUARY 2023 – JANUARY 2024
Contract Professor:
Instructor for Master Course of *"Advanced Computer Architectures"* for Computer Science and Engineering Master Degree with 231 students. Credits: 5 CFU. Course Evaluation: 3.2/4.

Politecnico di Milano FEBRUARY 2023 – JANUARY 2024
Contract Professor:
Instructor for Bachelor Course of *"Informatica e Elementi Di Informatica Medica"*, equivalent to CS101 in C, for Bionegineering Bachelor Degree with 191 students. Credits: 7 CFU. Course Evaluation: 2.6/4.

Passion in Action – Politecnico di Milano OCTOBER 2022 – FEBRUARY 2024
Instructor:
Instructor for for four editions (spring '24, fall'23, spring '23, fall'22). Course of *"FPGA101"* which introduces the Field-Programmable Gate Arrays technology along with primary design flows and system design methodologies.

CPS Summer School JULY 2022 – SEPTEMBER 2022
Creative Lab Instructor:
Instructor of the Creative Lab of the Cyber-Physical Systems (CS) Summer School held in Pula (ITA). Lab director M. D. Santambrogio.

Politecnico di Milano – CEFRIEL JULY 2022 – JULY 2022
Instructor:
Course of *"Hardware and Accelerators: FPGAs for AI"* part of the "Artificial Intelligence And Machine Learning Applications" Specialization degree ("Master universitario di primo livello")

Passion in Action – Politecnico di Milano FEB 2022 – JUNE 2022
Lecturer and Tutor:
Course of *"FPGA Academy"* held by Prof Marco D. Santambrogio

Passion in Action – Politecnico di Milano OCT 2020 – JUNE 2021
Lecturer and Tutor:
Course of *"FPGA Academy"* held by Prof Marco D. Santambrogio

Passion in Action – Politecnico di Milano FEB 2020 – JUNE 2020
Lecturer and Tutor:
Course of *"FPGA Academy"* held by Prof Marco D. Santambrogio

OTHER TEACHING ACTIVITIES

Politecnico di Milano FEB 2024 – JUNE 2024
Teaching Assistant:
Master Course of *"Advanced Computer Architectures"* held by Prof Christian Pilato with around 170 Students. Credits: 5 CFU. Course Evaluation: n.a.

Politecnico di Milano FEB 2023 – JUNE 2023
Teaching Assistant:
Master Course of *"Advanced Computer Architectures"* held by myself with 231 Students. Credits: 5 CFU. Course Evaluation: 3.2/4

Politecnico di Milano FEB 2022 – JUNE 2022
Teaching Assistant:
Master Course of *"Advanced Computer Architectures"* held by Prof Donatella Sciuto with 181 Students. Credits: 5 CFU. Course Evaluation: 3.1/4

Politecnico di Milano FEB 2021 – JUNE 2021
Teaching Assistant:
Master Course of *"Advanced Computer Architectures"* held by Prof Marco D. Santambrogio with 251 Students. Credits: 5 CFU. Course Evaluation: 3.3/4

Politecnico di Milano FEB 2020 – JUNE 2020
Teaching Assistant:
Master Course of *"Advanced Computer Architectures"* held by Prof Marco D. Santambrogio with 222 Students. Credits: 5 CFU. Course Evaluation: 2.7/4

MAR 2020 – APR 202
Volunteering Assistant:
Volunteering to assist online teaching and online Polimi degrees during first Covid-19 lockdown

Politecnico di Milano FEB 2019 – JUNE 2019
Teaching Assistant:
Master Course of *"Digital Systems and Design Methodologies 1"* held by Prof Fabrizio Ferrandi with 80 Students. Credits: 5 CFU. Course Evaluation: 2.7/4

Politecnico di Milano FEB 2019 – JUNE 2019
Project Tutor:
Bachelor Course of *"Prova Finale (Progetto di Reti Logiche) "* held by Prof Gianluca Palermo with 147 Students. Credits: 1 CFU. Course Evaluation: n.a.

Politecnico di Milano MAY 2022 – MAY 2022
Tutor:
"Hackathon: Informatica e Elementi di Informatica Medica" of the Bachelor course held by Prof Marco D. Santambrogio.

Politecnico di Milano OCT 2019 – NOV 2019
Tutor:
"Hackathon: Hack the NECSTCamp" held by Prof Marco D. Santambrogio.

PARTICIPATION/RESPONSIBILITY FOR FUNDED PROJECTS

EVEREST EU H2020 PROJECT, CONTRACT NO. 957269 SEPT 2022 – SEP 2023
Task Leader:
Task Leader of the T3.1 Data allocation and storage in the WP3 Data management techniques of the European Project *"EVEREST: dEsign enVironmEnt foR Extreme-Scale big data analyTics on heterogeneous platforms"* by the Horizon 2020 EU Research & Innovation programme. Total budget ~5M€, Politecnico di Milano funding amount: ~650K€.

Huawei Technologies, Zurich Research Center FEB 2022 – JAN 2023
Participant, Post-Doc Researcher :
PostDoc researcher for the project entitled *"Templated Spatial Architectures"* by Huawei.

NVIDIA SEP 2021 – SEP 2022
Participant, Post-Doc Researcher :
PostDoc researcher on a grant entitled *"Software-programmable Domain Specific Architectures for Regular Expressions"* consisting on hardware donations from NVIDIA.

Tecnosens SEP 2019 – SEP 2020
Automation Infrastructure Lead and Hardware Developer :
Researcher for the project *"Software-defined hardware pipeline enabling high performance OCR over heterogeneous image streams"* from Tecnosens.

JULIGHT S.r.l. DEC 2018 – DEC 2019
Support:
Support researcher for the project *"Individuazione angolo vivo/morto"* from JULIGHT

IN2IT MAR 2018 – JULY 2018
Contributor to WP5:
Participant and contributor to the IN2IT: "*Internationalization by Innovative Technology European Project*", with a special focus on WP5, where, in particular, I have successfully completed the summary meeting "Development and exploitation of academy-industry/community cooperation" held in Kingston, (UK).

OTHER EXPERIENCE

NECSTLab - Politecnico di Milano CURRENT FROM JAN 2020
AMD-Xilinx License and Toolchain manager:
Responsible for managing, distributing, and handling AMD-Xilinx toolchains and licenses across machines and users.

NECSTLab - Politecnico di Milano CURRENT FROM JAN 2020
Coursera Sessions for FPGA-based courses:
Responsible for managing coursera private sessions linked to FPGA-based courses.

COMPUTER SKILLS

PROG. LANG.	C, C++, Python, Java
HW LANG.	VHDL, Verilog, HLS, Chisel
OS USAGE	Ubuntu, CentOS, Windows, RHEL, MacOS
TOOL	AMD-Xilinx Vivado, Vitis, PYNQ
VARIOUS	LaTeX, Git, Makefile, bash, TCL, Asana, Trello, JIRA

SCIENTIFIC PRODUCTIVITY AND IMPACT METRICS

Scientific Productivity: **29** publications (**24** entries on Scopus, **35** co-authors according to Scopus):

- Author/Co-author of **10** journal papers, of which **7** top-ranked **Q1** journal papers based on SCIMAGO (including IEEE Transactions on Parallel and Distributed Systems, IEEE Transactions on Emerging Topics in Computing, ACM Computing Surveys, ACM Transactions on Reconfigurable Technology and Systems, IEEE Journal of Biomedical and Health Informatics, Journal of Parallel and Distributed Computing);
- Author/Co-author of **18** scientific publications on peer-reviewed conferences among DAC, CGO, FPGA;
- Author/Co-author of **1** Book Chapter.

Publication Impact:

Based on Google Scholar:	h-index **9**	citations **172**
Based on Scopus:	h-index **7**	citations **102**

Data collected on the 21st February 2024, for Journals: Scimago; Conferences: CORE, GGS, and Conference Ranks.
Research Profiles:

- Dblp: https://dblp.org/pid/224/1533.html
- Scholar: https://scholar.google.it/citations?user=Y0VnEtkAAAAJ
- Orcid: https://orcid.org/0000-0002-5834-0812
- Scopus: https://www.scopus.com/authid/detail.uri?authorId=57203552342

JOURNAL PUBLICATIONS

[J1] Alberto Zeni, Emanuele Del Sozzo, Eleonora D'Arnese, **Davide Conficconi**, and Marco D Santambrogio. "Starlight: A Kernel Optimizer for GPU Processing". In: *Journal of Parallel and Distributed Computing* (2023). *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking 2022*: **Q1**, SJR 1.16 [Scimago]. DOI: 10.1016/j.jpdc.2023.104832.

[J2] Emanuele Del Sozzo, **Davide Conficconi**, and Kentaro Sano. "Across Time and Space: Senju's Approach for Scaling Iterative Stencil Loop Accelerators on Single and Multiple FPGAs". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* (2023). *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking 2022*: **Q1**, SJR 0.91 [Scimago]. DOI: 10.1145/3634920.

[J3] Giuseppe Sorrentino, Marco Venere, **Davide Conficconi**, Eleonora D'Arnese, and Marco D Santambrogio. "HEPHAESTUS: Codesigning and Automating 3D Image Registration on Reconfigurable Architectures". In: *ACM Transactions on Embedded Computing Systems (TECS)* 22.5s (2023). *Citations*: 1 [Google Scholar], 1 [Scopus]. *Ranking 2022*: **Q2**, SJR 0.8 [Scimago]. ISSN: 1539-9087. DOI: 10.1145/3607928. URL: https://doi.org/10.1145/3607928.

[J4] Raffaele Berzoini, Eleonora D'Arnese, **Davide Conficconi**, and Marco D. Santambrogio. "NERONE: the Fast Way to Efficiently Execute Your Deep Learning Algorithm at the Edge". In: *IEEE Journal of Biomedical and Health Informatics (J-BHI)* (2023). *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking 2022*: **Q1**, SJR 1.67 [Scimago]., pp. 1–9. DOI: 10.1109/JBHI.2023.3296142. URL: https://doi.org/10.1109/JBHI.2023.3296142.

[J5] Eleonora D'Arnese, **Davide Conficconi**, Emanuele Del Sozzo, Luigi Fusco, Donatella Sciuto, and Marco D Santambrogio. "Faber: a Hardware/Soft-ware Toolchain for Image Registration". In: *IEEE Transactions on Parallel and Distributed Systems* (2022). *Citations*: 5 [Google Scholar], 3 [Scopus]. *Ranking 2022*: **Q1**, SJR 1.89 [Scimago]. DOI: 10.1109/TPDS.2022.3218898. URL: https://doi.org/10.1109/TPDS.2022.3218898.

[J6] Emanuele Del Sozzo, **Davide Conficconi**, Alberto Zeni, Mirko Salaris, Donatella Sciuto, and Marco Domenico Santambrogio. "Pushing the Level of Abstraction of Digital System Design: a Survey on How to Program FPGAs". In: *ACM Computing Surveys (CSUR)* (2022). *Citations*: 25 [Google Scholar], 19 [Scopus]. *Ranking 2022*: **Q1**, SJR 4.46 [Scimago]. DOI: 10.1145/3532989. URL: https://doi.org/10.1145/3532989.

[J7] **Davide Conficconi**, Emanuele Del Sozzo, Filippo Carloni, Alessandro Comodi, Alberto Scolari, and Marco Domenico Santambrogio. "An Energy-Efficient Domain-Specific Architecture for Regular Expressions". In: *IEEE Transactions on Emerging Topics in Computing* (2022). *Citations*: 10 [Google Scholar], 2 [Scopus]. *Ranking 2022*: **Q1**, SJR 1.35 [Scimago]. DOI: 10.1109/TETC.2022.3157948. URL: https://doi.org/10.1109/TETC.2022.3157948.

[J8] Daniele Parravicini, **Davide Conficconi**, Emanuele Del Sozzo, Christian Pilato, and Marco D Santambrogio. "CICERO: A Domain-Specific Architecture for Efficient Regular Expression Matching". In: *ACM Transactions on Embedded Computing Systems (TECS)* 20.5s (2021). *Citations*: 11 [Google Scholar], 8 [Scopus]. *Ranking 2021*: **Q2**, SJR 0.75 [Scimago]., pp. 1–24. DOI: 10.1145/3476982. URL: https://doi.org/10.1145/3476982.

[J9] Enrico Reggiani, Emanuele Del Sozzo, **Davide Conficconi**, Giuseppe Natale, Carlo Moroni, and Marco D Santambrogio. "Enhancing the scalability of multi-fpga stencil computations via highly optimized hdl components". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 14.3 (2021). *Citations*: 13 [Google Scholar], 11 [Scopus]. *Ranking 2021*: **Q1**, SJR 0.88 [Scimago]., pp. 1–33. DOI: 10.1145/3461478. URL: https://doi.org/10.1145/3461478.

[J10] Yaman Umuroglu, **Davide Conficconi**, Lahiru Rasnayake, Thomas B Preusser, and Magnus Själander. "Optimizing bit-serial matrix multiplication for reconfigurable computing". In: *ACM Transactions on Reconfigurable Technology and Systems (TRETS)* 12.3 (2019). *Citations*: 24 [Google Scholar], 10 [Scopus]. *Ranking 2019*: **Q3**, SJR 0.26 [Scimago]., pp. 1–24. DOI: 10.1145/3337929. URL: https://doi.org/10.1145/3337929.

CONFERENCE PUBLICATIONS

[C1] Filippo Carloni, **Davide Conficconi**, and Marco D Santambrogio. "ALVEARE: a Domain-Specific Framework for Regular Expressions". In: *2024 61th ACM/IEEE Design Automation Conference (DAC)*. *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking*: A [CORE], GGS: 2, CORE:A, LiveSHINE:A++, MA:A++ [GSS]. 2024, pp. 1–7. DOI: Accepted--To--Appear.

[C2] Luisa Cicolini, Filippo Carloni, Marco D Santambrogio, and **Davide Conficconi**. "One Automaton To Rule Them All: Beyond Multiple Regular Expressions Execution". In: *IEEE/ACM International Symposium on Code Generation and Optimization*. *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking*: A [CORE], GGS: 2, CORE:A, LiveSHINE:A+, MA:A- [GSS]. 2024, pp. 1–15. DOI: https://doi.org/10.1109/CGO57630.2024.10444810.

[C3] Giuseppe Sorrentino, Marco Venere, Eleonora D'Arnese, **Davide Conficconi**, Isabella Poles, and Marco D Santambrogio. "ATHENA: a GPU-based Framework for Biomedical 3D Rigid Image Registration". In: *IEEE Biomedical Circuits and Systems Conference (BioCAS)*. *Citations*: 1 [Google Scholar], 0 [Scopus]. *Ranking*: MA: C [GSS]. 2023, pp. 1–5. DOI: https://hdl.handle.net/11311/1252460.

[C4] Beatrice Branchini, **Davide Conficconi**, Donatella Sciuto, and Marco Santambrogio. "The Hitchhiker's Guide to FPGA-Accelerated Quantum Error Correction". In: *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*. Vol. 02. *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking*: Qualis: B3 [Conference Ranks]. 2023, pp. 338–339. DOI: 10.1109/QCE57702.2023.10271. URL: https://doi.org/10.1109/QCE57702.2023.10271.

[C5] Marco Venere, Giuseppe Sorrentino, Beatrice Branchini, **Davide Conficconi**, Elisabetta Di Nitto, Donatella Sciuto, and Marco Santambrogio. "On the Design and Characterization of Set Packing Problem on Quantum Annealers". In: *IEEE EUROCON 2023 International Conference on Smart Technologies*. *Citations*: 1 [Google Scholar], 0 [Scopus]. *Ranking*: Qualis: B3 [Conference Ranks]. 2023, pp. 695–700. DOI: 10.1109/EUROCON56442.2023.10199096. URL: https://doi.org/10.1109/EUROCON56442.2023.10199096.

[C6] Beatrice Branchini, **Davide Conficconi**, Francesco Peverelli, Donatella Sciuto, and Marco Santambrogio. "A Bird's Eye View on Quantum Computing: Current and Future Trends". In: *IEEE EUROCON 2023 International Conference on Smart Technologies*. *Citations*: 1 [Google Scholar], 1 [Scopus]. *Ranking*: Qualis: B3 [Conference Ranks]. 2023, pp. 689–694. DOI: 10.1109/EUROCON56442.2023.10198957. URL: https://doi.org/10.1109/EUROCON56442.2023.10198957.

[C7] Roberto Alessandro Bertolini, Filippo Carloni, and **Davide Conficconi**. "Co-designing an FPGA-Accelerated Encryption Library With PYNQ: The Pynqrypt Case Study". In: *IEEE EUROCON 2023 International Conference on Smart Technologies*. *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking*: Qualis: B3 [Conference Ranks]. 2023, pp. 683–688. DOI: 10.1109/EUROCON56442.2023.10198938. URL: https://doi.org/10.1109/EUROCON56442.2023.10198938.

[C8] Filippo Carloni, Leonardo Panseri, **Davide Conficconi**, Mattia Sironi, and Marco D. Santambrogio. "Enabling Efficient Regular Expression Matching at the Edge through Domain-Specific Architectures". In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. *Citations*: 0 [Google Scholar], 0 [Scopus]. *Ranking*: MA: C [GSS], Qualis: B3 [Conference Ranks]. 2023, pp. 71–74. DOI: 10.1109/IPDPSW59300.2023.00023. URL: https://doi.org/10.1109/IPDPSW59300.2023.00023.

[C9] Filippo Carloni, **Davide Conficconi**, Ilaria Moschetto, and Marco D. Santambrogio. "YARB: A Methodology to Characterize Regular Expression Matching on Heterogeneous Systems". In: *2023 IEEE International Symposium on Circuits and Systems (ISCAS)*. *Citations*: 1 [Google Scholar], 1 [Scopus]. *Ranking*: GGS Class: 2, Rating: A-, LiveSHINE: A+, MA: A- [GSS]. 2023, pp. 1–

5. DOI: 10.1109/ISCAS46773.2023.10181547. URL: https://doi.org/10.1109/ISCAS46773.2023.10181547.

[C10] Emanuele Del Sozzo, **Davide Conficconi**, Marco D. Santambrogio, and Kentaro Sano. "Senju: A Framework for the Design of Highly Parallel FPGA-based Iterative Stencil Loop Accelerators". In: *Proceedings of the 2023 ACM/SIGDA International Symposium on Field-Programmable Gate Arrays*. *Citations*: 1 [Google Scholar], 0 [Scopus]. *Ranking*: Class 2, Rating A, LiveSHINE: A, MA: A [GSS]. 2023, p. 233. DOI: 10.1145/3543622.3573170. URL: https://doi.org/10.1145/3543622.3573170.

[C11] Francesco Peverelli, **Davide Conficconi**, Davide Basilio Bartolini, Alberto Scolari, and Marco D. Santambrogio. "Characterizing Molecular Dynamics Simulation on Commodity Platforms". In: *2022 IEEE International Symposium on Workload Characterization (IISWC)*. *Acceptance Rate*: 47.9% (23/48). *Citations*: 2 [Google Scholar], 2 [Scopus]. *Ranking*: Rating A-, MA: A- [GSS]. 2022, pp. 65–78. DOI: 10.1109/IISWC55918.2022.00016. URL: https://doi.org/10.1109/IISWC55918.2022.00016.

[C12] Raffaele Berzoini, Eleonora D'Arnese, and **Davide Conficconi**. "On How to Push Efficient Medical Semantic Segmentation to the Edge: the SENECA approach". In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. *Citations*: 3 [Google Scholar], 1 [Scopus]. *Ranking*: MA: C [GSS], Qualis: B3 [Conference Ranks]. 2022. DOI: 10.1109/IPDPSW55747.2022.00027. URL: https://doi.org/10.1109/IPDPSW55747.2022.00027.

[C13] Daniele Paletti, Francesco Peverelli, and **Davide Conficconi**. "Online Learning RTL Synthesis for Automated Design Space Exploration". In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. *Citations*: 2 [Google Scholar], 2 [Scopus]. *Ranking*: MA: C [GSS], Qualis: B3 [Conference Ranks]. 2022. DOI: 10.1109/IPDPSW55747.2022.00021. URL: https://doi.org/10.1109/IPDPSW55747.2022.00021.

[C14] Eleonora D'Arnese, Emanuele Del Sozzo, **Davide Conficconi**, and Marco D Santambrogio. "Exploiting Heterogeneous Architectures for Rigid Image Registration". In: *IEEE Biomedical Circuits and Systems Conference (BioCAS)*. *Citations*: 5 [Google Scholar], 3 [Scopus]. *Ranking*: MA: C [GSS]. 2021, pp. 1–5. DOI: 10.1109/BioCAS49922.2021.9645026. URL: https://doi.org/10.1109/BioCAS49922.2021.9645026.

[C15] Giulia Gerometta, **Davide Conficconi**, and Marco Domenico Santambrogio. "On How FPGAs are Changing the Computer Security Panorama: An Educational Survey". In: *IEEE 6th International Forum on Research and Technology for Society and Industry (RTSI)*. *Citations*: 2 [Google Scholar], 1 [Scopus]. *Ranking*: n.a. . 2021, pp. 80–85. DOI: 10.1109/RTSI50628.2021.9597337. URL: https://doi.org/10.1109/RTSI50628.2021.9597337.

[C16] Daniele Paletti, **Davide Conficconi**, and Marco D Santambrogio. "Dovado: An Open-Source Design Space Exploration Framework". In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. *Citations*: 7 [Google Scholar], 3 [Scopus]. *Ranking*: MA: C [GSS], Qualis: B3 [Conference Ranks]. 2021, pp. 128–135. DOI: 10.1109/IPDPSW52791.2021.000271. URL: https://doi.org/10.1109/IPDPSW52791.2021.00027.

[C17] **Davide Conficconi**, Eleonora D'Arnese, Emanuele Del Sozzo, Donatella Sciuto, and Marco D Santambrogio. "A Framework for Customizable FPGA-based Image Registration Accelerators". In: *ACM/SIGDA International Symposium on Field-Programmable Gate Arrays*. *Acceptance Rate*: 23.2% (26/111). *Citations*: 17 [Google Scholar], 14 [Scopus]. *Ranking*: GGS Class 2, Rating A, LiveSHINE: A, MA: A [GSS]. 2021, pp. 251–261. DOI: 10.1145/3431920.3439291. URL: https://doi.org/10.1145/3431920.3439291.

[C18] Lorenzo Di Tucci, **Davide Conficconi**, Alessandro Comodi, Steven Hofmeyr, David Donofrio, and Marco D Santambrogio. "A parallel, energy efficient hardware architecture for the merAligner on FPGA using Chisel HCL". In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. *Citations*: 10 [Google Scholar], 7 [Scopus]. *Ranking*: MA: C [GSS], Qualis: B3 [Conference Ranks]. 2018, pp. 214–217. DOI: 10.1109/IPDPSW.2018.00041. URL: https://doi.org/10.1109/IPDPSW.2018.00041.

[C19] Alessandro Comodi, **Davide Conficconi**, Alberto Scolari, and Marco D Santambrogio. "TiReX: Tiled regular expression matching architecture". In: *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. *Citations*: 16 [Google Scholar], 13 [Scopus]. *Ranking*: MA: C [GSS], Qualis: B3 [Conference Ranks]. 2018, pp. 131–137. DOI: 10.1109/IPDPSW.2018.00028. URL: https://doi.org/10.1109/IPDPSW.2018.00028.

BOOK CHAPTERS

[B1] Eleonora D'Arnese, **Davide Conficconi**, Marco Domenico Santambrogio, and Donatella Sciuto. "Reconfigurable architectures: the shift from general systems to domain specific solutions". In: *Emerging Computing: From Devices to Systems. Looking Beyond Moore and Von Neumann*. Ed. by Anupam Chattopadhyay Mohamed M. Sabry Aly. *Citations*: 13 [Google Scholar], 0 [Scopus]. *Ranking* : n.a. . Singapore: Springer Nature Singapore, 2023, pp. 435–456. ISBN: 978-981-16-7487-7. DOI: 10.1007/978-981-16-7487-7_14. URL: https://doi.org/10.1007/978-981-16-7487-7_14.

AWARDS

2023 **ACM Artifacts** *of Available, Reusable, Validated & Reproduced* for [C2]

2023 **Co-Supervisor of a Winner Team of AMD Open Hardware Design Contest**
Heterogeneous Highly Integrated Systems for Image Registration, G. Sorrentino, P. Galfano

2023 **HiPEAC Summer School ACACES Grant**

2023 **IEEE Artifacts** of *Open Research Objects*, *Research Objects Reviewed*, *Results Reproduced* for [C8]

2023 **Young People Programme** award at DATE'23

2022 **IEEE Artifacts** of *Open Research Objects*, i.e., research artifact open-source, for [C11]

2022 **Co-Supervisor of a Winner Team of Xilinx Open Hardware Design Contest**
A Journey to the center of the 3D Space, G. Sorrentino, M. Venere

2022 **IBM Best PhD Intern Presentation** at Cloud & AI Systems Research (CAISR) department Intern Symposium July '22

2022 **First Place IEEE 2022 Lance Stafford Larson Award** on [C17]

2022 **ASPLOS 2022 Student Travel Award**

2021 **NVIDIA Academic Hardware Grant Program** *"Software-programmable Domain Specific Architectures for Regular Expressions"*

2021 **ACM Artifacts** *of Available, Reusable, Reproduced* for [C17]

2019 **Finalist of Xilinx Open Hardware Design Contest**
IRON Image RegistratiOn oN FPGA

2018 **First Prize Shark Tank** Internal Interns Competition at Xilinx Dublin with "*Deep Breath: Measuring Precisly the Air Pollution*"

2017 **Finalist of Xilinx Open Hardware Design Contest**
TiReX: Tiled Regular eXpression matching architecture

2017 2018 **Scholarship for High Merits at PoliMi**
Partial tuition waiver for high academic performance

2017 **Category Winners Xilinx PYNQ Hackathon at PoliMi**
Smart glove for remote controlling

CONFERENCE ACTIVITIES

Committee and Organization

ARTIFACT EVALUATION CHAIR	RAW '24 RAW '23
PROGRAM COMMITTEE	FPL '24, FCCM '24, GLSVLSI '24, HEART '24, SAC'24 - EMBS Track , RAW '24 ISPA '23, GLSVLSI '23, RAW '23 RAW '22
ARTIFACT EVALUATION COMMITTEE	CGO '24 MICRO '23, ASPLOS '23, CGO '23, PLDI '23 ASPLOS '22
TUTORIAL ORGANIZER	"Faber: a Hardware/Software Toolchain for Image Registration" at EUROCON '23
SESSION CHAIR	"Computer science, quantum solutions and digital twins" at EUROCON '23 RAW '23, RAW '21
VOLUNTEER	DATE '21, DATE '19
BOOTCAMP ORGANIZER	Chisel Bootcamp at ASAP '18

External Reviewer

REVIEWER	ISCAS '24, FPL '23, AFRICON '23 EUROCON '23 ISPA '22,HEART '22, 2xDATE'22, DAC '22 DATE '21, FCCM '20, 2xICS '20
SUBREVIEWER	2xDAC '21, RAID '21, 2xDAC '20, DATE '20, RAW '20, FPL '20, CODES+ISSS '19, ReConFig '19

Attendance

2024	HPCA'24, CGO'24, CC'24
2023	DATE'23, RAW'23
2022	ASPLOS '22, NOPE'22, DATE'22, GTC '22, HEART'22, IPDPS'22, FPT'22
2021	H^2RC '21, ESWEEK '21, GTC '21, Xilinx Adapt'21, DATE'21, FPGA'21, RAW'21, ISCA'21, ISVLSI'21
2020	HOTI '20, FPL'20, DAC'20, Xilinx Adapt'20
2019	DATE'19, ICS'19, PhD Workshop on "Next-Generation Cloud Infrastructure" MSR Cambridge
2018	RAW'18
2017	RAW'17

JOURNAL ACTIVITIES

APRIL 2023 - MAY 2024

Guest Editor:

Guest Editor for the Special Issue on Secure and Efficient Distributed Computation for Emerging Systems on the Edge (SUNRISE) on Elsevier Journal of Parallel and Distributed Computing (JPDC): Opened July 2023.

SEPT. 2020 - CURRENT

Reviewer:

Revising 20+ manuscripts across different journals:
IEEE Transactions on Computer Aided Design of Integrated Circuits and Systems (TCAD): 2024, 2022;
ACM Transactions on Reconfigurable Technology and Systems (TRETS): 2024, 2023;
IEEE Access: 2024, 2022, 2021;
IEEE Transactions on Emerging Topics in Computing (TETC): 2023;
IEEE Embedded Systems Letters (ESL): 2023;
IEEE Transactions on Very Large Scale Integration (VLSI) Systems: 2023;
ACM Transactions on Embedded Computing Systems (TECS): 2023, 2022;
IEEE IT Professional (ITPro): 2023;
Elsevier Computes and Security (COSE):2021;
Elsevier Microprocessors and Microsystems (MICPRO):2021 ;

IEEE Transaction on Parallel and Distributed Sytems (TPDS): 2021, 2020;
ACM Transactions on Design Automation of Electronic Systems (TODAES): 2021;

EXTERNAL EXPERT ACTIVITIES

PHD REFREE	Thesis "Graph-based techniques and strategies for diagnosis and characterization of neurodegenerative diseases ", 2023, by Laura Hernández Lorenzo, Universidad Complutense de Madrid, Advisors: J. Ayala, J. Antem.
PROPOSAL REVIEW	Swiss National Science Foundation (SNSF) 2023
AWARD REVIEWERS	IEEE CS Larson Award reviewer for 2023 fall cycle

ASSOCIATION AND VOLUNTEER

AVIS	Blood Donor Volunteer since 2017
IEEE	Member (2017)
ACM	Member (2020)
HIPEAC	Affiliated Member (2023)
IEEE	Computer Society (2017)
ACM	SIGARCG Member (2024)
IEEE	CS Technical Community on Computer Architecture

IEEE	Circuits and Systems
IEEE	Women In Enegineering
IEEE	Volunteer as Larson Award Chair
POLIMI	Open-Day Volunteer 23; 17–20
VOLUNTEER	Different local fairs for catering

RESEARCH TOOLS

IMFANT	One Automaton To Rule Them All: Beyond Multiple Regular Expressions Execution [C2] https://github.com/necst/iMFAnt
ATHENA	A GPU-based Framework for Biomedical 3D Rigid Image Registration [C3] https://github.com/necst/athena
HEPHAESTUS	Codesigning and Automating 3D Image Registration on Reconfigurable Architectures [J3] https://github.com/necst/hephaestus
NERONE	NERONE : the Fast Way to Efficiently Execute Your Deep Learning Algorithm at the Edge [J4] https://github.com/necst/NERONE
QUANTUM BENCHMARK.	On the Design and Characterization of Set Packing Problem on Quantum Annealers [C5] https://github.com/necst/quantum_annealer_benchmarking
CICERO ON ARDUINO	Enabling Efficient Regular Expression Matching at the Edge through Domain-Specific Architectures [C8] https://github.com/necst/cicero-on-vidor4000
YARB	YARB: a Methodology to Characterize Regular Expression Matching on Heterogeneous Systems [C9] https://github.com/necst/yarb
FABER FPGA	Faber: a Hardware/Software Toolchain for Image Registration [J5] https://github.com/necst/faber_fpga
MD BENCH.	A Benchmark Suite for characterizing Molecular Dynamics (MD) simulation on commodity platforms [C11] https://github.com/necst/lammps-benchmarks
FPGA-PROG.	A list of ways to design and program the FPGAs according [J6] https://github.com/emanueledelsozzo/awesome-fpga-programming
SENECA	Deploying efficient medical semantic segmentation models on edge devices [C12] https://github.com/necst/seneca
MOVADO	A Black-Box fitness function approx. library [C13] https://github.com/necst/movado
CFP_ZOO	A cloudFPGA project with domain-specific accelerators for the hybrid multi-cloud era. https://github.com/cloudFPGA/cFp_Zoo
FABER GPU	Exploiting heterogeneous architectures for rigid Image Registration [C14]. https://github.com/necst/faber_biocas
CICERO	A domain specific architecture for Regular Expression matching on FPGA [J8]. https://github.com/necst/cicero
DOVADO	A framework for RTL-based Design Space Exploration [C16] https://github.com/necst/dovado
XLNX	A template repository for Xilinx FPGAs designs https://github.com/necst/xlnx-project-template
IRON	A framework for customizable FPGA-based Image Registration accelerators [C17]. https://github.com/necst/iron
BISMO	A bit-serial DSA for few-bits matrix multiplications [J10]. https://github.com/EECS-NTNU/bismo
CHISEL-SDX	A Chisel wrapper for SDAccel integration [C18] https://github.com/necst/sdaccel_chisel_integration

COMMUNICATION SKILLS

TALK

2023 "From Domain-Specific to Quantum Computing: the Role of Reconfigurable Systems", at IEEE EUROCON 2023, July 2023, Torino, (ITA)
2023 "Enabling Efficient Regular Expression Matching at the Edge through Domain-Specific Architectures", at RAW '23

2022 "Characterizing the CloudFPGA System", at CAISR Interns Syposium, IBM Research, Zurich (CH)

2021 "On the Role of Reconfigurable Systems in Domain Specific Computing", Colloquia Doctoralia at PoliMi, Milan (ITA).
2021 "A Framework for Customizable FPGA-based Image Registration Accelerators", at H^2RC '21
2021 "Dovado: An Open-Source Design Space Exploration Framework", at RAW '21
2021 "A Framework for Customizable FPGA-based Image Registration Accelerators", at FPGA '21

2018 "Deep Breath", Shark Tank at Xilinx, Dublin (IE).
2018 "TiReX: Tiled Regular eXpression matching architecture", at RAW '18

INVITED TALK

2023 "Intro to the NECSTLab and the NECST Research Line Fair Event (NRLFE)", at several companies (e.g., Bosh, Edutech District, Bending Spoons, Amazon) in presence and online, NRLFE, Politecnico di Milano, Milan (IT)
2023 "Domain-Specific Computing Research Line at NECSTLab", at Boston University, Boston, MA (USA).
2023 "Faber: Hardware/Software Toolchain for Image Registration", at Northeastern University, Boston, MA (USA).

2022 "On the Role of Reconfigurable Systems in Domain Specific Computing", NECST Friday Talk at PoliMi, Milan (ITA).
2022 "Intro to the NECSTLab and the NECST Research Line Fair Event (NRLFE)", at several companies (e.g., Mindway, ABE Elettronica, Xlogic, Techedge) in presence and online, NRLFE, Politecnico di Milano, Milan (IT)
2022 "Domain-Specific Computing Research Line at NECSTLab", at Huawei Research at Polimi, Milan (ITA)
2022 "Faber: Hardware/Software Toolchain for Image Registration", at Northwestern University, IL (USA).
2022 "Domain-Specific Computing Research Line at NECSTLab", at Northwestern University, IL (USA).
2022 "Software-programmable Domain-Specific Architectures for Regular Expressions", virtually held to NVIDIA DPU team.
2022 "Faber: Hardware/Software Toolchain for Image Registration", at University of Illinois at Chicago, IL (USA).
2022 "Domain-Specific Computing Research Line at NECSTLab", at University of Illinois at Chicago, IL (USA).
2022 "Faber: Hardware/Software Toolchain for Image Registration", virtually held to international institutions: Microsoft Research, Lawrence Berkeley National Laboratory, Xilinx Inc.

2020 "DRACO: Domain-specific Reconfigurable Architecture Computer Organization", PoliMi, Milan (ITA).

2019 "DRACO: Domain-specific Reconfigurable Architecture Computer Organization", Open Networking Foundation (ONF), Menlo Park, CA (USA).
2019 "DRACO: Domain-specific Reconfigurable Architecture Computer Organization", Lawrence Berkeley National Laboratory, Berkeley, CA (USA).
2019 "DRACO: Domain-specific Reconfigurable Architecture Computer Organization", Xilinx, San Josè, CA (USA).
2019 "TiReX: Tiled Regular eXpression matching architecture", Xilinx, San Josè, CA (USA).
2019 "DRACO: Domain-specific Reconfigurable Architecture Computer Organization", Xilinx, San Josè, CA (USA).

2018 "TiReX: Tiled Regular eXpression matching architecture", Xilinx, San Josè, CA (USA).

2017 "TiReX: Tiled Regular eXpression matching architecture", Microsoft Research at PoliMi, Milan (ITA)
2017 "TiReX: Tiled Regular eXpression matching architecture", Xilinx, San Josè, CA (USA).

POSTER

2023 "Hardware/Software Acceleration of Heuristic-based Image Registration on Heterogeneous Systems", 19th International Summer School on Advanced Computer Arch. and Compilation for High-performance Embedded Systems '23, Fiuggi (ITA)
2023 "On the Role of Reconfigurable Systems in Domain Specific Computing", DATE'23 PhD Forum, Antwerp (BE)

2019 "DRACO: Domain specific Reconfigurable Architecture Computer Organization", PhD Workshop on Next-Generation Cloud Infrastructure '19, MSR, Cambridge

2018 "TiReX: Tiled Regular eXpression matching architecture", Xilinx, San Josè, CA (USA).

SEMINARS

2019 "On how to develop from software to hardware for the ZYNQ technology and the Ultra96", Tecnosens and Imavis at PoliMi, Milan (ITA).

LECTURES

CPS Creative Lab '23, '22

Advanced Computer Architectures '23 till '20
FPGA101 '23 fall, '23 spring, '22 fall
Hardware and Accelerators: FPGAs for AI'22
FPGAcademy '22 spring-'21-'20
Digital System and Design Methodologies 1 '19

COURSES Participant "Pitching your research to key audiences" by H. Gustaffon, NTNU, 2020

Participant "Embracing Diversity" 1+2 by IN2IT Platform, 2018

Participant "English for Internationalization" 1+2 by IN2IT Platform, 2018

Participant "Startup 101" by S. Notargiacomo, PoliMi, 2017

WORKSHOP Participant"The 5 Chairs of Leadership" and "Assertiveness", L. Evans, Milan, May'23
Participant "Active Learning and Feedback for soft skills", METID, Milan, Nov'19
Participant "Presentation skills and storyline building", BCG, Milan, Nov'19
Participant "PhD Workshop on Next-Gen. Cloud Infrastructure", MSR, Cambridge, Nov'19

Participant "Design Thinking", Bosch, Feb'19

LANGUAGES Italian Native, English Fluent, German Beginner

COMMISSION OF TRUST

2023 Nov-Dec, **External referee Ph.D. thesis** in Computer Science and Engineering, Universidad Complutense de Madrid.

2023 October, **Master thesis defense committee member** in Computer Science and Engineering, 10 students, Politecnico di Milano.

2023 May, **Master thesis defense committee member** in Computer Science and Engineering, 11 students, Politecnico di Milano.

MASTER THESIS ADVISOR

2023 OCTOBER **"One Automaton To Rule Them All: Enabling Multiple Regular Expressions Execution"**
Student: *L. Cicolini*
Co-Advisor: *F. Carloni*

2023 JULY **"HEPHAESTUS: an FPGA-based Framework for 3D Image Registration'**
Student: *G. Sorrentino*
Co-Advisor: *E. D'Arnese*

2023 APRIL **"On the Feasibility of Optimizing ML-Based Intrusion Detection for CAN on Real-world Hardware Platforms"**
Student: *E. Massaro*
Co-Advisor: *S. Longari*

MASTER THESIS CO-ADVISOR

2022 OCTOBER **"AutoREX: a Methodology for Regular Expressions Benchmarking on Heterogeneous Architectures"**
Student: *I. Moschetto*
Advisor: *M. D. Santambrogio*
Co-Advisor: *F. Caloni*

2022 APRIL **"YBoost: a framework to accelerate YARA rules pattern matching using FPGAs"**
Student: *A. Furlan*
Advisor: *S. Zanero*
Co-Advisor: *M. Carminati, M. Polino*

2021 OCTOBER **"Alveare: A Novel Mixed HW-SW Framework for Efficient Execution of Regular Expressions"**
Student: *F. Carloni*
Advisor: *M. D. Santambrogio*